

Ricardo Barrios

Full Stack Developer and Systems Engineering Student that loves to build experiences to help others.

(305) 426-4649 | dejesusbg5@gmail.com | Santa Marta, Colombia

github.com/dejesusbg | linkedin.com/in/dejesusbg | dejesusbg.netlify.app

Education

Bachelor in Systems Engineering · **Universidad del Magdalena** Expected July 2026

- **Coursework:** Artificial Intelligence, Web Development, Operating Systems, Microservices, UI Design
- **Achievements:** 4.2 / 5.0 CGPA

Experience

UX Research Intern · **Universidad del Magdalena** Nov 2025 — Feb 2026

- **Will lead usability and accessibility research** for academic systems, enhancing course registration and scheduling for **40+ undergraduate programs and 16k+ students**, from degree coordinator perspectives.
- **Will design and conduct user research**, gathering data from diverse stakeholders, through the execution of moderated interviews, surveys, and usability tests (remote and in-person).
- **Will deliver evidence-based redesign proposals** to research directors, improving system user experience through quantitative and qualitative insights.

Web Developer · **INGNOVA Specialized Engineering** Feb 2025 — June 2025

- **Launched the company's first corporate website** (rdingnova.com.co) showcasing 9+ specialized civil engineering services, developed with Next.js, TypeScript, React, and Tailwind CSS.
- **Enabled 2-minute content updates** for non-technical staff, streamlining site maintenance, by integrating Decap CMS with Cloudinary CDN and automating SEO metadata.
- **Achieved top Google search rankings and 99+ Lighthouse scores** across Performance, Accessibility, and SEO, delivering high visibility, responsiveness, and user-friendly experience.

Projects

Truenavi · **Blind-First Navigation App** github.com/dejesusbg/truenavi

- **Led a 4-person team to create a voice-controlled navigation app** to assist visually impaired users in urban environments with hands-free interaction, using React Native, Node.js, Express, and MongoDB.
- **Achieved 95%+ speech recognition accuracy** and sub-400 ms route calculations via graph algorithms.
- **Cut database latency** by 40% with REST and NoSQL geospatial optimizations.

Mangifera · **AI-Powered Mango Classifier** github.com/dejesusbg/mangifera

- **Built and tuned computer vision classifiers on 2,714 images**; applied augmentation, feature engineering and cross-validation to raise accuracy from 70% to 95%+.
- Explored models such as **deep neural networks, random forests, and linear regression models** and used hyperparameter search to maximize generalization for agricultural decision support.

Skills

Programming Languages : TypeScript, Python, JavaScript, HTML/CSS, PHP, Java

Libraries & Frameworks : React, Next.js, React Native, Node.js, Express, Tailwind, Spring Boot, PostgreSQL, MongoDB, Redis, TensorFlow, scikit-learn, pandas

Tools & Platforms : Docker, Git, GitHub, Netlify, Figma, Power BI, Lighthouse, Keycloak